INJECTIVITY OF THE UNIT F -> F_* F^{-1}F FOR A SURJECTIVE PROPER MORPHISM, WITH IMAGE THE EQUALIZER

THE STACKS PROJECT AUTHORS

Editorial note. The counted statement is the author's own lemma environment labelled lemma-surjective-proper (source lines 365-373); the proof environment beginning on the line after the following blank line (source line 375) is its own proof and closes at source line 403.

Editorial note. The proof cites four results, all outside the chapter (Properties of Spaces, Cohomology of Spaces, Topologies, Etale Cohomology); each is declared as a tagged source pointer, and no step the counted argument needs is left to them: the four pointers are used exactly as the author uses them, as named inputs.

Editorial note. The opening paragraph of the surrounding text ('The proper base change theorem for algebraic spaces follows from ...', source lines 360-363) is not part of the counted statement or proof and is not retained.

Editorial note. The retained preamble is the chapter's own (sources/stacks-spaces-more-cohomology/preamble.tex, taken with its real line count), a verbatim excerpt of the native Xy-pic setup and theorem declarations; no macro was invented or redefined.

Editorial scope. One original result of the Stacks Project's chapter Cohomology of Algebraic Spaces, counted once: for a surjective proper morphism f the unit F -> f_*f^{-1}F is injective with image the equalizer.

Allowed standard prerequisites (editorial).

- Pullback of a sheaf along a morphism and its stalks. For a morphism f : Y -> X of algebraic spaces the unit F -> f_*f^{-1}F, whose stalk at a geometric point y over x is F_x via the canonical isomorphism (Properties of Spaces, declared as a tagged source pointer).. It computes the composite at a geometric point in the injectivity step (source lines 381-387).
- Etale locality of the equalizer statement. The statement that the image is the equalizer is local on X for the etale topology, so one may assume Y is an affine scheme.. It is the reduction at source lines 390-391.
- Chow's lemma for algebraic spaces. Every algebraic space Y admits a surjective proper morphism Z -> Y with Z a scheme (Cohomology of Spaces, declared as a tagged source pointer).. It produces the scheme Z at source lines 394-395.
- Surjective proper morphisms of schemes are ph coverings, and the scheme case. A surjective proper morphism of schemes is a ph covering (Topologies), and for schemes the equalizer statement is known (Etale Cohomology,

Date: stacks-project:spaces-more-cohomology.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

declared as tagged source pointers).. They transfer the result from Z to Y at source lines 397-402.

Source reference. Original lemma 04K2: sources/stacks-chow/tags.txt, source lines 5958–5958. Cross-chapter pointer whose target carries the canonical tag 04K2 in the project tag list; declared so the reference resolves inside the assembled package, while every step the counted argument uses remains inside the retained spans.

Source reference. Original lemma 089J: sources/stacks-chow/tags.txt, source lines 10499–10499. Cross-chapter pointer whose target carries the canonical tag 089J in the project tag list; declared so the reference resolves inside the assembled package, while every step the counted argument uses remains inside the retained spans.

Source reference. Original lemma 0DES: sources/stacks-chow/tags.txt, source lines 16812–16812. Cross-chapter pointer whose target carries the canonical tag 0DES in the project tag list; declared so the reference resolves inside the assembled package, while every step the counted argument uses remains inside the retained spans.

Source reference. Original lemma 0DDW: sources/stacks-chow/tags.txt, source lines 16781–16781. Cross-chapter pointer whose target carries the canonical tag 0DDW in the project tag list; declared so the reference resolves inside the assembled package, while every step the counted argument uses remains inside the retained spans.

Lemma 0.1. *Let S be a scheme. Let $f : Y \to X$ be a surjective proper morphism of algebraic spaces over S. Let $\mathcal{F}$ be a sheaf on $X_{étale}$. Then $\mathcal{F} \to f_*f^{-1}\mathcal{F}$ is injective with image the equalizer of the two maps $f_*f^{-1}\mathcal{F} \to g_*g^{-1}\mathcal{F}$ where g is the structure morphism $g : Y \times_X Y \to X$.*

Proof. For any surjective morphism $f : Y \to X$ of algebraic spaces over S, the map $\mathcal{F} \to f_*f^{-1}\mathcal{F}$ is injective. Namely, if $\overline{x}$ is a geometric point of X, then we choose a geometric point $\overline{y}$ of Y lying over $\overline{x}$ and we consider

$$\mathcal{F}_{\overline{x}} \to (f_*f^{-1}\mathcal{F})_{\overline{x}} \to (f^{-1}\mathcal{F})_{\overline{y}} = \mathcal{F}_{\overline{x}}$$

See Properties of Spaces, Lemma 04K2 for the last equality.

The second statement is local on X in the étale topology, hence we may and do assume Y is an affine scheme.

Choose a surjective proper morphism $Z \to Y$ where Z is a scheme, see Cohomology of Spaces, Lemma 089J. The result for $Z \to X$ implies the result for $Y \to X$. Since $Z \to X$ is a surjective proper morphism of schemes and hence a ph covering (Topologies, Lemma 0DES) the result for $Z \to X$ follows from Étale Cohomology, Lemma 0DDW (in fact it is in some sense equivalent to this lemma). □